# Matrices & Vectors

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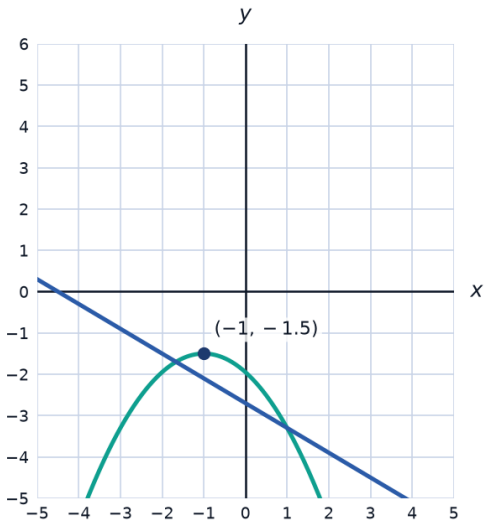
## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** **[Foundational]** • **Target: Addition and scalar multiplication**

Let $A$ be the matrix with rows $\langle 2, 1\rangle$ and $\langle 3, -1\rangle$, and $B$ be the matrix with rows $\langle -1, 4\rangle$ and $\langle 0, 2\rangle$. Find $A + B$ and $3A$.

**2** **[Foundational]** • **Target: Addition and scalar multiplication**

Using the same $A$ and $B$, find $A - 2B$.




**3** **[Foundational]** • **Target: Matrix multiplication**

Using $A$ (rows $\langle 2, 1\rangle$, $\langle 3, -1\rangle$) and $B$ (rows $\langle -1, 4\rangle$, $\langle 0, 2\rangle$) from above, find $AB$.

**4** **[Foundational]** • **Target: Matrix multiplication**

Let $A$ have rows $\langle 2, 1\rangle$ and $\langle 3, -1\rangle$, and let $\mathbf{v} = \langle 2, -3\rangle$. Find $A\mathbf{v}$.

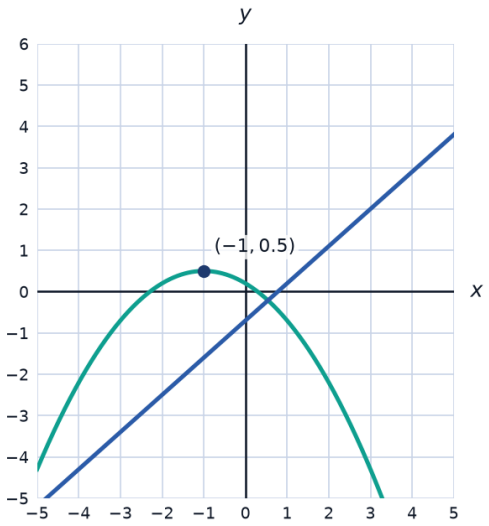
**5** **[Foundational]** • **Target: Matrix multiplication**

Explain, without computing $BA$, why $AB$ and $BA$ need not be equal for $2 \times 2$ matrices.

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**6** [Foundational] • Target: Determinant and inverse

Find $\det A$ for the matrix $A$ with rows $\langle 2, 1\rangle$ and $\langle 3, -1\rangle$, and state whether $A$ is invertible.




**7** [Foundational] • Target: Determinant and inverse

Find $A^{-1}$ for the matrix $A$ with rows $\langle 2, 1\rangle$ and $\langle 3, -1\rangle$.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Determinant and inverse

Without computing it, explain why the matrix with rows $\langle 4, 2\rangle$ and $\langle 6, 3\rangle$ has no inverse.

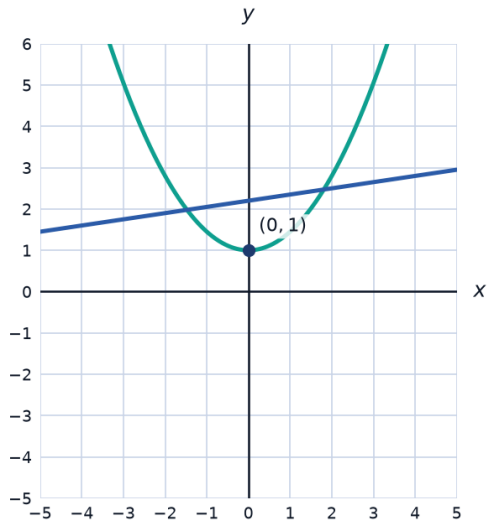
**9** [Intermediate] • Target: Solving systems with the inverse matrix

Solve the system $3x - y = 8$ and $x + 2y = 5$ using $\mathbf{x} = M^{-1}\mathbf{b}$.

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**10** [Intermediate] • Target: Vector magnitude and dot product

What does $\mathbf{u} \cdot \mathbf{w} = 0$ tell you about $\mathbf{u}$ and $\mathbf{w}$ from the previous question?




**11** [Intermediate] • Target: Angle between vectors

Find the angle between $\mathbf{p} = \langle 1, 0 \rangle$ and $\mathbf{q} = \langle 1, 1 \rangle$.

**12** [Intermediate] • Target: Angle between vectors

Find the angle between $\langle 6, -8 \rangle$ and $\langle 3, 4 \rangle$, rounded to the nearest tenth of a degree.

**13** [Intermediate] • Target: Addition and scalar multiplication

Given 3D vectors $\mathbf{u} = \langle 2, -3, 6 \rangle$ and $\mathbf{v} = \langle 1, 4, 2 \rangle$: (a) Find the unit vector in the direction of $\mathbf{u}$. (b) Calculate the angle $\theta$ between $\mathbf{u}$ and $\mathbf{v}$ to the nearest degree.

**14** [Intermediate] • Target: Matrix multiplication

Find all real values of the constant $k$ for which the matrix $M = (k + 2, 4; 3, k - 2)$ is singular (has no inverse).

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

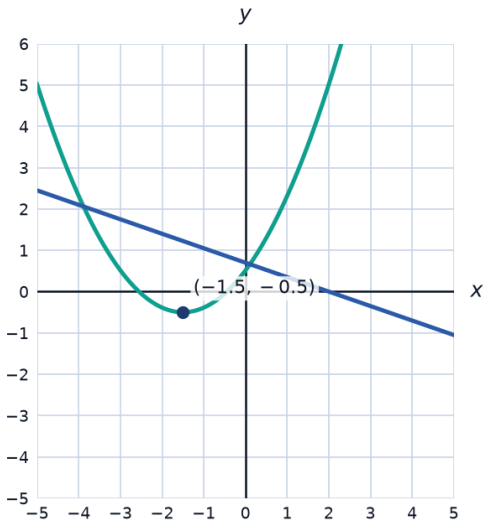
**15** [Challenging] • Target: Determinant and inverse • Cross-Lesson: 1.01 Complex Numbers

Find the magnitude and direction angle $\theta$ (in degrees, $0^\circ \le \theta < 360^\circ$) of the vector $\mathbf{v} = \langle -5, 5\sqrt{3} \rangle$.

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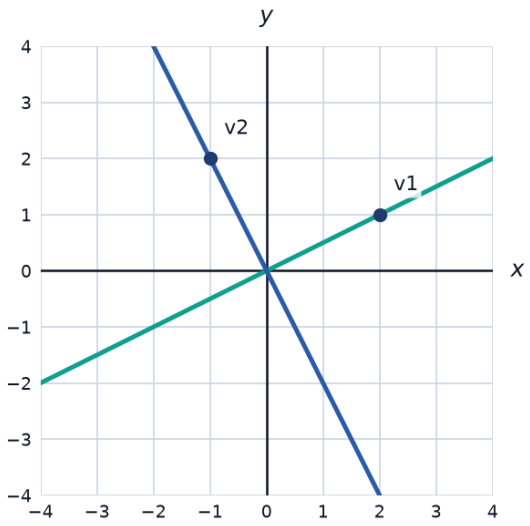
**16** [Challenging] • Target: Solving systems with the inverse matrix • Cross-Lesson: 3.01 Sequences & Series

Find the angle $\theta$ between the vectors $\mathbf{a} = \langle 1, 2, 2 \rangle$ and $\mathbf{b} = \langle 2, -2, 1 \rangle$.




**17** [Challenging] • Target: Vector magnitude and dot product • Cross-Lesson: 4.01 Advanced Functions & Equations

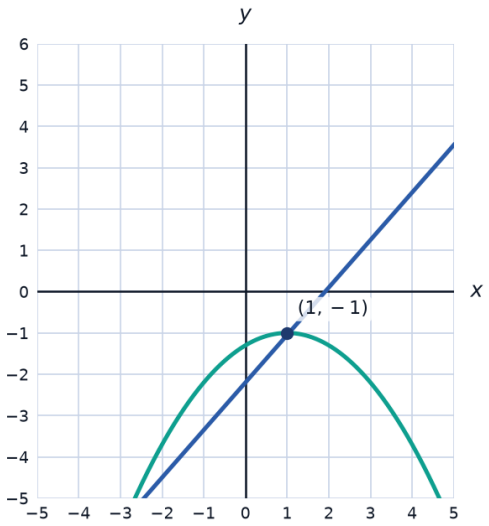
Find the inverse of the matrix $A = (4, 7; 1, 2)$.




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**18** [Challenging] • Target: Angle between vectors • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

Compute the matrix product $AB$ where $A = (1, 3; -2, 4)$ and $B = (2, 0; 5, -1)$.




**19** [Challenging] • Target: Addition and scalar multiplication • Cross-Lesson: 6.01 Limits

Vectors $\mathbf{a} = \langle 4, k \rangle$ and $\mathbf{b} = \langle 6, -8 \rangle$ are perpendicular. Find the value of $k$, and compute the projection of $\mathbf{c} = \langle 10, 5 \rangle$ onto $\mathbf{b}$.

**20** [Challenging] • Target: Matrix multiplication • Cross-Lesson: 1.01 Complex Numbers

Let $A = (3, 1; 5, 2)$. Find $A^{-1}$ and use it to solve the linear system: $3x + y = 7$ $5x + 2y = 12$

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Addition and scalar multiplication

For $\mathbf{u} = \langle 20, 21 \rangle$ and $\mathbf{v} = \langle -5, -12 \rangle$, find $\cos\theta$ where $\theta$ is the angle between them, using $\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$.

**a.** $\frac{352}{377}$

**b.** $-\frac{352}{377}$

**c.** $-\frac{27}{29}$

**d.** $-\frac{176}{21}$

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**22** [Foundational] • Target: Matrix multiplication

Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in the coordinate plane such that $|\mathbf{u}| = 5$, $|\mathbf{v}| = 10$, and the angle $\theta$ between them is obtuse ($90^\circ < \theta < 180^\circ$). If the minimum value of $|t\mathbf{u} + \mathbf{v}|$ over all real numbers $t$ is equal to $6$, what is the value of the dot product $\mathbf{u} \cdot \mathbf{v}$?

**a.** -50

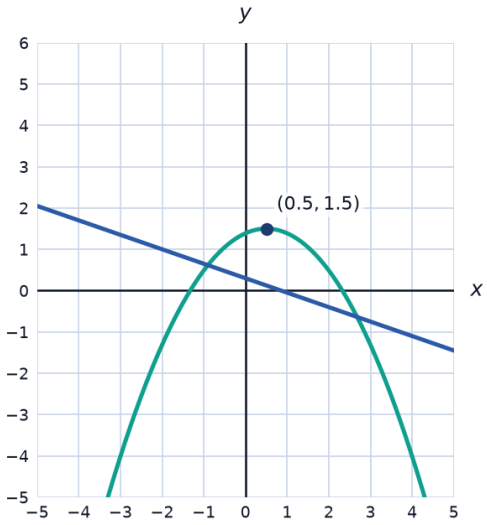
**b.** -40

**c.** -30

**d.** 40

**23** [Foundational] • Target: Determinant and inverse

In the standard $(x, y)$-coordinate plane, vectors $\mathbf{u} = \langle a, b \rangle$ and $\mathbf{v} = \langle c, d \rangle$ have magnitudes $||\mathbf{u}|| = 13$ and $||\mathbf{v}|| = 5$. If their dot product is $\mathbf{u} \cdot \mathbf{v} = 25$, and the matrix $M = [a, b; c, d]$ has $\det(M) > 0$, what is the value of $\det(M)$?




**a.** 25

**b.** 40

**c.** 60

**d.** 65

**24** [Foundational] • Target: Solving systems with the inverse matrix

For all real values of $\theta$, the matrix product $[\cos\theta, \sin\theta][6, 2; 2, 3][\cos\theta; \sin\theta]$ evaluates to a $1 \times 1$ matrix $[k]$. What is the maximum possible value of $k$?

**a.** 6

**b.** 7

**c.** 9

**d.** 14

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**25** [Foundational] • Target: Vector magnitude and dot product

Matrices $A$ and $B$ are defined as $A = [2, k; -1, 3]$ and $B = [1, 4; 2, k]$, where $k$ is a real number. If the determinant of the matrix sum $A + B$ is equal to $17$, what is the value of $k$? (Note: The determinant of a $2 \times 2$ matrix $[a, b; c, d]$ is defined as $ad - bc$.)

**a.** 1

**b.** 2

**c.** 6

**d.** $\frac{19}{2}$

**26** [Foundational] • Target: Angle between vectors

For a $2 \times 2$ matrix $[a, b; c, d]$, the determinant is defined as $ad - bc$. Given the matrices $A = [x, 2; -1, 3]$ and $B = [1, -4; 3, x]$, the determinant of the matrix $A + B$ is equal to $19$. If $x > 0$, what is the value of $x$?

**a.** 2

**b.** $\frac{5}{4}$

**c.** 4

**d.** 6

**27** [Foundational] • Target: Addition and scalar multiplication

In the standard coordinate plane, vector $\mathbf{u} = \langle 2, -1 \rangle$ and vector $\mathbf{v} = \langle k, 5 \rangle$, where $k$ is a real constant. If the magnitude of the vector $\mathbf{u} + \mathbf{v}$ is equal to 5, which of the following is a possible value of $k$?

**a.** -5

**b.** -1

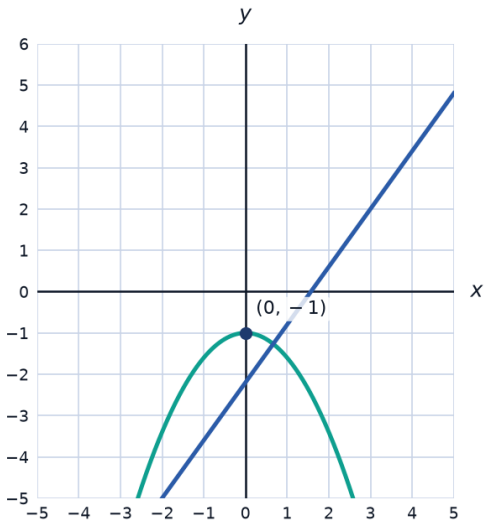
**c.** 3

**d.** 7

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**28** [Foundational] • Target: Matrix multiplication

What is the value of the determinant of the matrix $[7, -3; 2, 4]$?




**a.** 34

**b.** 22

**c.** -29

**d.** -34

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Determinant and inverse

Find the angle $\theta$ between $\mathbf{u} = \langle -6, 0 \rangle$ and $\mathbf{v} = \langle 6, -1 \rangle$, to the nearest $0.1^\circ$.

**a.** $170.5^\circ$

**b.** $9.5^\circ$

**c.** $180.5^\circ$

**d.** $165.5^\circ$

**30** [Intermediate] • Target: Solving systems with the inverse matrix

Solve using the inverse-matrix method: $\{-5x + 3y = 28;\ -5x + 0y = 25$.

**a.** x=-5,\ y=1

**b.** x=1,\ y=-5

**c.** x=-4,\ y=1

**d.** x=-5,\ y=-1

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**31** [Intermediate] • Target: Vector magnitude and dot product

Given the vectors $\mathbf{u} = \langle 3, -4 \rangle$ and $\mathbf{v} = \langle -2, 5 \rangle$, what is the vector $2\mathbf{u} + \mathbf{v}$?

**a.** $\langle 4, -3 \rangle$

**b.** $\langle 1, 1 \rangle$

**c.** $\langle 4, 1 \rangle$

**d.** $\langle 8, -13 \rangle$

**32** [Intermediate] • Target: Angle between vectors

If matrix $A = [3, -2; 0, 4]$ and matrix $B = [-1, 5; 2, -3]$, what is the matrix $2A + B$?

**a.** $[5, 1; 2, 5]$

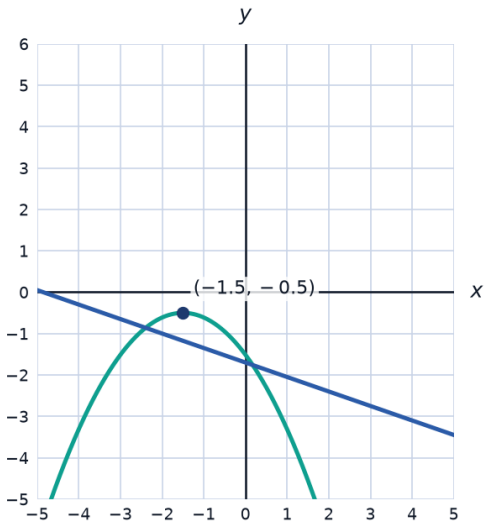
**b.** $[2, 3; 2, 1]$

**c.** $[5, 3; 2, 1]$

**d.** $[7, -9; -2, 11]$

**33** [Intermediate] • Target: Addition and scalar multiplication

Given matrix $A = (2, -1; 3, 3)$, what is the scalar product $3A$?




**a.** $(5, 2; 6, 6)$

**b.** $(6, -1; 3, 9)$

**c.** $(6, -3; 9, 9)$

**d.** $(6, 3; 9, 10)$

**34** [Intermediate] • Target: Matrix multiplication

A transformation maps point $A(1, 2)$ to $A'(-4, 2)$ and point $B(3, 5)$ to $B'(-2, 5)$. This transformation can be represented by matrix multiplication $(x\ y)' = M(x\ y) + (a\ b)$. Find the values of $a$ and $b$.

**a.** a = -5, b = 0

**b.** a = 5, b = 0

**c.** a = 0, b = -5

**d.** a = 0, b = 5

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**35** [Intermediate] • Target: Determinant and inverse

Consider a linear transformation $T$ represented by the matrix $A = (2, 1\ 1, 3)$. Let $P = (x; y)$ be a point in the plane. The transformation maps $P$ to $P'$. If the area of the region transformed by $T$ from a unit square in the $xy$-plane is $A$, and the determinant of $A$ is $\det(A)$, find the value of $\det(A)$ if the area of the transformed region is $5$.

**a.** 5

**b.** 2

**c.** 1

**d.** 10

**36** [Intermediate] • Target: Solving systems with the inverse matrix

Consider the matrix $A = (2, 0; 0, 3)$. When the matrix $A$ is applied to the vector $\mathbf{v} = (x; y)$, the resulting vector is $A\mathbf{v} = (6; 9)$. What is the original vector $\mathbf{v}$?

**a.** $(3; 3)$

**b.** $(1; 2)$

**c.** $(6; 3)$

**d.** $(12; 27)$

**37** [Intermediate] • Target: Vector magnitude and dot product

Compute the dot product $\mathbf{u} \cdot \mathbf{v}$ for $\mathbf{u} = \langle 2, 3 \rangle$ and $\mathbf{v} = \langle 2, 5 \rangle$.

**a.** 19

**b.** 22

**c.** 16

**d.** 14

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Angle between vectors • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

What is the magnitude of the 2D vector $\mathbf{v} = \langle 3, -4 \rangle$?

**a.** 5

**b.** 12

**c.** 10

**d.** 7

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**39** [Challenging] • Target: Addition and scalar multiplication • Cross-Lesson: 6.01 Limits

In $\Delta ABC$, the side lengths are $AB = 5$ and $BC = 4$, and $\angle B$ is an acute angle. If $\cos(2B) = -\frac{31}{32}$, what is the length of side $AC$?

**a.** $6$

**b.** $\sqrt{46}$

**c.** $\sqrt{41}$

**d.** $\sqrt{31}$

**40** [Challenging] • Target: Matrix multiplication • Cross-Lesson: 1.01 Complex Numbers

In $\Delta ABC$, the lengths of sides $AB$ and $BC$ are $5$ and $8$, respectively, and the measure of $\angle B$ is $60^\circ$. What is the length of the altitude from vertex $B$ to side $AC$?

**a.** $\frac{20\sqrt{3}}{7}$

**b.** $\frac{40\sqrt{3}}{7}$

**c.** $\frac{10\sqrt{3}}{7}$

**d.** $\frac{20}{7}$

**41** [Challenging] • Target: Determinant and inverse • Cross-Lesson: 3.01 Sequences & Series

In $\Delta ABC$, the length of side $AB$ is $x$, the length of side $BC$ is $x + 3$, and the length of side $AC$ is $7$. If $\angle B = 60^\circ$, what is the value of $x$?

**a.** $5$

**b.** $8$

**c.** $4$

**d.** $6$

**42** [Challenging] • Target: Solving systems with the inverse matrix • Cross-Lesson: 4.01 Advanced Functions & Equations

In $\Delta ABC$, $A = 64^\circ$, $a = 18$, $b = 9$. Find angle $B$ to the nearest tenth (assume $B$ acute).

**a.** $26.7^\circ$

**b.** $28.7^\circ$

**c.** $89.3^\circ$

**d.** $23.7^\circ$

**43** [Challenging] • Target: Vector magnitude and dot product • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

In $\Delta ABC$, $a = 13$, $b = 9$, included angle $C = 60^\circ$. Find $c$ to the nearest tenth.

**a.** $11.5$

**b.** $12.5$

**c.** $10.8$

**d.** $22$

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**44** [Challenging] • Target: Angle between vectors • Cross-Lesson: 6.01 Limits

In $\Delta ABC$, $a = 15$, $b = 7$, included angle $C = 60^\circ$. Find $c$.

**a.** 13

**b.** 14

**c.** 9

**d.** 17

**45** [Challenging] • Target: Addition and scalar multiplication • Cross-Lesson: 1.01 Complex Numbers

A linear transformation in the 2D plane is represented by the matrix $M = (a, b; c, d)$. This transformation maps the point $P(1, 2)$ to $P'(-3, 7)$ and the point $Q(3, 1)$ to $Q'(5, 1)$. Determine the values of $a$, $b$, $c$, and $d$, and then calculate the sum $a + b + c + d$.

**a.** 9

**b.** 11

**c.** 7

**d.** 5

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